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A Final Report On

Maximizing Incremental Store Visits Under a Limited SMS Budget: Customer Segmentation and Uplift Modeling for SMS Targeting

In partial fulfillment of the requirements for the degree of Bachelor of Science in Data Science, with a minor in Marketing.

Submitted By:
Nuria Castellsague Kharbouch

Course: Data Science Capstone
Instructor: Matthew Powell

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Executive Summary

When SMS budgets are limited, the right business question is not which customers are most likely to visit stores, but which customers are most likely to visit because they receive a message. This project addresses that incrementality problem using the BIGTARGET customer-level marketing dataset, which includes both a treatment indicator for SMS recipients and an outcome indicator for store visit behavior.

The analysis began by cleaning and standardizing customer behavioral variables, reducing dimensionality with principal components analysis (PCA), and clustering customers into four behavior-based segments using k-means. Segment-level uplift was then estimated by comparing treated and control visit rates within each segment. To test whether the discovered segments reflected real behavior structure rather than arbitrary clustering, a 4-class random forest model was trained to predict segment membership. Finally, Segment 2 and 3 were combined into a binary persuadables class and modeled directly to create a more operational targeting tool.

The strongest result is that Segment 2 produced the largest and most reliable incremental lift from SMS at about 0.80 percentage points, followed by Segment 3 at about 0.59 percentage points. Segment 1 had the highest overall visit rate, but its confidence interval crossed zero, which suggests many of these customers would have visited anyway. Segment 4 was the largest group and still showed a positive lift, but its baseline visit rates were very low, making it a weak strategic target relative to the persuadable groups.

The business recommendation is to move away from broad, undifferentiated SMS outreach and toward targeted deployment based on persuadability. Under a limited budget, the company should prioritize customers in Segment 2 first and Segment 3 second, while avoiding unnecessary spend on customers who are likely to visit without prompting or who remain weak responders even after contact.

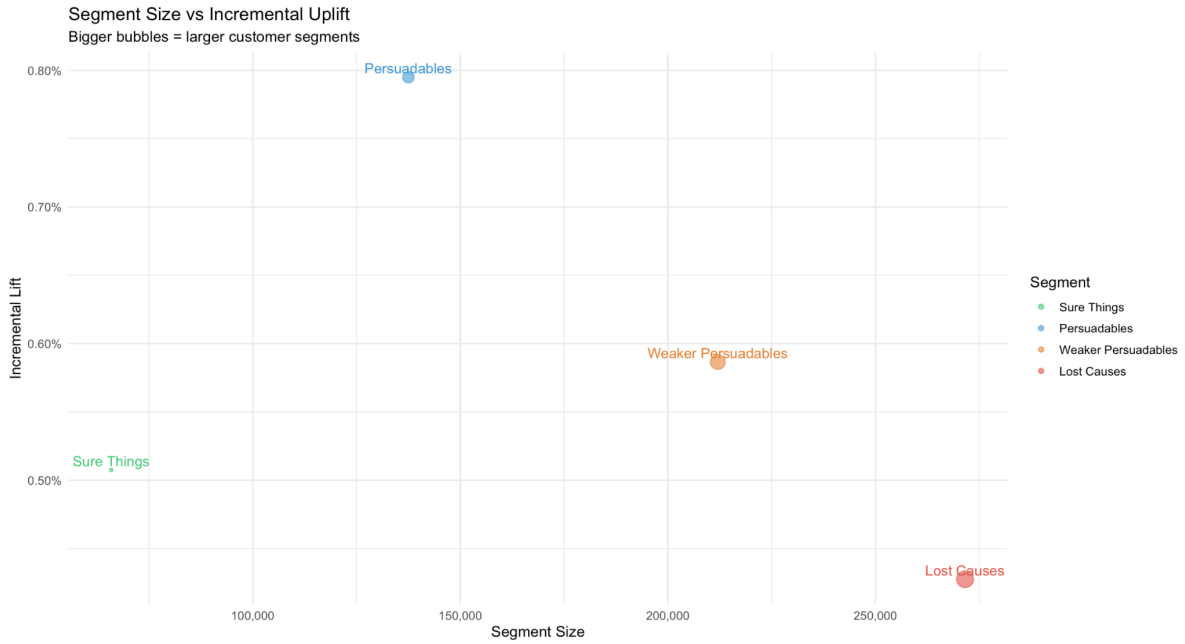

Figure 1: Segment size versus incremental uplift. Larger bubbles indicate larger customer groups; higher position indicates greater SMS lift.

Introduction

Businesses with limited marketing budgets cannot afford to treat all customers the same. In practice, broad message volume often looks efficient, but it can be wasteful when many customers would have purchased anyway, some will not respond regardless of treatment, and a few may even respond negatively to outreach. The real challenge in customer relationship management is not simply predicting outcomes, but deciding which customers should be contacted because the contact itself is expected to change behavior.

This project is motivated by a classic marketing framework of four customer types. Sure Things are customers who visit or purchase whether contacted or not. Persuadables are customers who become more likely to visit when contacted. Lost Causes are unlikely to respond either way. Do-Not-Disturbs are customers who may be less likely to respond when contacted. From a targeting perspective, the ideal strategy is to identify persuadables while avoiding spending on the other groups.

That logic is especially important for smaller brands and budget-constrained campaigns, where messaging as many people as possible can be destructive rather than helpful. This report applies that framework to SMS targeting using customer-level transactional and behavioral data, with the aim of identifying where SMS produces truly incremental store visits.

Business Problem and Project Objective

The motivating question for this capstone is: If SMS budget is limited, how can customers be targeted to maximize incremental store visits relative to sending no SMS? This is a customer relationship management (CRM) and resource-allocation problem. Maximizing total visits is not the same as maximizing incremental visits, because some customers will visit even without a message. Contacting those customers still incurs campaign cost but adds little to no marginal value.

The primary objective of the project is therefore to identify customer groups whose behavior changes positively because of SMS contact. The secondary objective is to build a predictive framework that can be used to target future campaigns more efficiently. Treatment in this study is whether a customer received an SMS message, and the outcome is whether that customer visited or purchased in the attribution period.

This project deliberately emphasizes value and incremental lift rather than accuracy alone. A highly accurate model that predicts visit propensity without distinguishing incremental response may still produce poor business decisions. By contrast, a model that identifies who is persuadable supports a much stronger targeting strategy.

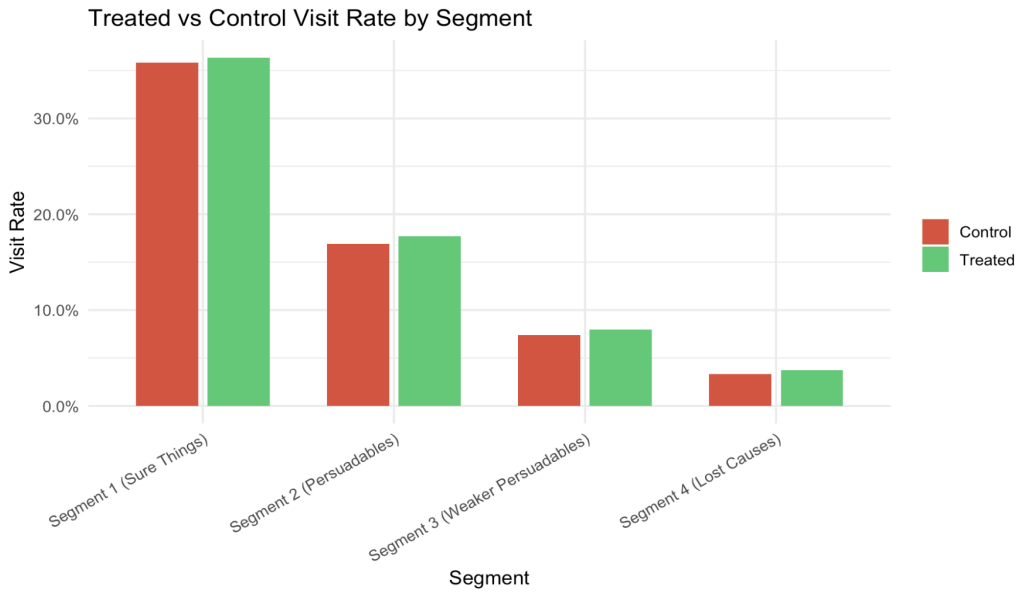

Figure 2: Treated versus control visit rate by segment. The treated-control gap represents the incremental effect of SMS within each segment.

Dataset and Data Availability

The dataset used in this project is the BIGTARGET hackathon dataset from the LENTA + Microsoft challenge hosted on Kaggle. It was explicitly designed to support better SMS targeting, which makes it especially appropriate for an incrementality-focused analysis. The unit of analysis is the individual customer, and the feature space is rich enough to support both behavioral segmentation and predictive modeling.

Available variables include demographic features such as age, household-related features such as number of children, and store-format indicators. More importantly, the dataset contains detailed behavioral measures including transaction counts by product group, receipt counts by product group, spending totals over different time windows, promotional and discount-related features, and behavioral variability measures such as changes in time between visits or discount depth. It also includes the two fields required for causal targeting: a treatment indicator showing whether an SMS was sent and an outcome indicator showing whether the customer visited during the attribution window.

These variables are well matched to the business problem because they capture both who the customer is and how the customer behaves. Frequency, recency, category preference, spending intensity, discount sensitivity, and regularity of visits are exactly the kinds of signals that help distinguish customers who are likely to respond incrementally from those who are not.

Data Cleaning and Preprocessing

The raw dataset required substantial cleaning before modeling. Character variables were trimmed, blank strings were converted to missing values, and customer ID fields were standardized. The treatment variable and the store-visit outcome were cleaned and forced into valid binary coding so that treated-versus-control comparisons would be interpretable. Numeric-like character variables were converted where needed, and infinite values were replaced with missing values.

Rule-based cleaning was also applied to feature families used in PCA and predictive modeling. Negative purchase counts, negative spending values, and invalid variation measures were treated as missing because they were not behaviorally meaningful. Share and rate variables were constrained to valid ranges. Basic demographic range checks were applied to variables such as age, number of children, and months from registration.

To make the data suitable for segmentation, variables with extreme zero-dominance and variables with very high missingness were filtered out from the PCA feature set. Remaining count and spending variables were transformed using log1p to reduce skewness and limit the influence of extreme outliers. Median imputation was used for remaining missing values, and features were standardized before PCA so that no single scale dominated the dimension-reduction step. These preprocessing decisions were necessary to produce stable, interpretable clusters and reliable downstream models.

Exploratory Data Analysis

Exploratory data analysis (EDA) showed that the dataset had the same characteristics often seen in large customer-behavior tables: very high dimensionality, heavy sparsity, and a small number of unusually active customers. Many purchase-related variables were extremely zero-heavy, several feature groups -- especially some variability measures -- had substantial missingness, and spending and count variables were strongly right-skewed.

These patterns made it clear that the raw variables were not suitable for direct clustering. Without filtering and transformation, a few large-value spending variables would dominate the analysis, while many near-constant variables would add noise without improving separation. The EDA, therefore, informed the final pipeline by motivating the use of zero-rate filtering, missingness filtering, log transformation, standardization, and PCA.

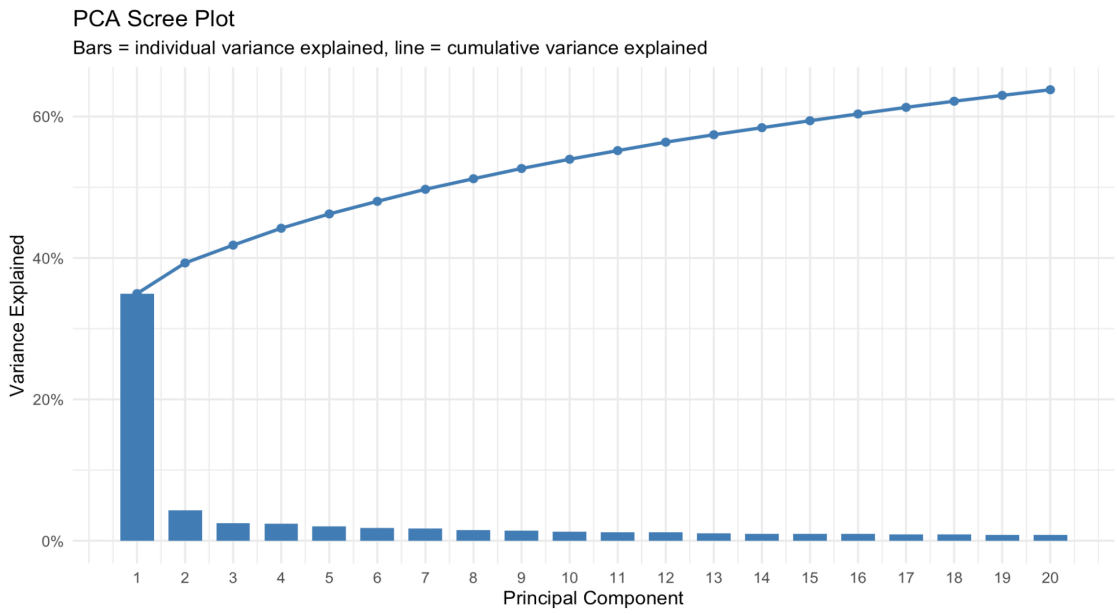

Figure 3: PCA scree plot. The first principal component explains the largest share of variation, followed by diminishing returns from later components.

Methodology Overview

The project used a multi-stage analytical workflow that moves from description to decision support. First, customer behavioral variables were cleaned, filtered, transformed, and standardized. Next, PCA was used to compress the high-dimensional behavioral space into a smaller set of informative components. K-means clustering was then applied in PCA space to discover four customer segments.

Once the segments were created, uplift was estimated within each segment by comparing treated and control visit rates. This step translated the behavioral segments into business-relevant customer types. After the segments were interpreted, a 4-class random forest model was trained to test whether segment membership was behaviorally stable and reproducible. Finally, because Segment 2 and Segment 3 both showed clear positive uplift, they were combined into a binary persuadable class and modeled directly as a practical targeting output.

Customer Segmentation Approach

Customer segmentation was performed with unsupervised learning because the goal was to discover natural grouping in historical customer behavior before assigning marketing labels. PCA reduced redundancy and noise across the many numeric features, and k-means clustering was applied using the first 10 principal components. A four-segment solution was selected because it produced behaviorally interpretable groups and aligned well with the classic marketing logic of Sure Things, Persuadables, and Lost Causes, with one intermediate persuadable group.

The final labels were not assigned by the clustering algorithm itself. Instead, the segments were interpreted by combining two kinds of evidence: their behavioral profiles and their uplift patterns. In this dataset, the most defensible empirical labels were Sure Things, Persuadables, Weaker Persuadables, and Lost Causes. Although the theoretical framework includes Do-Not-Disturbs, no segment displayed a clearly negative treated-control difference.

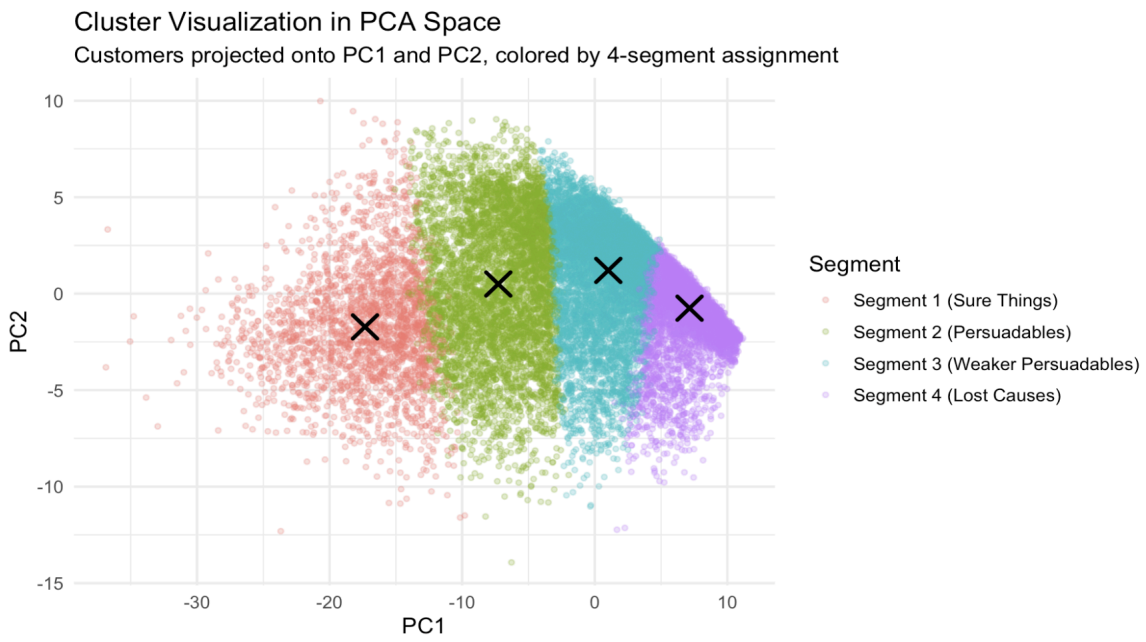

Figure 4: Cluster visualizations in PCA space. Customers are projected onto the first two principal components and colored by segment membership.

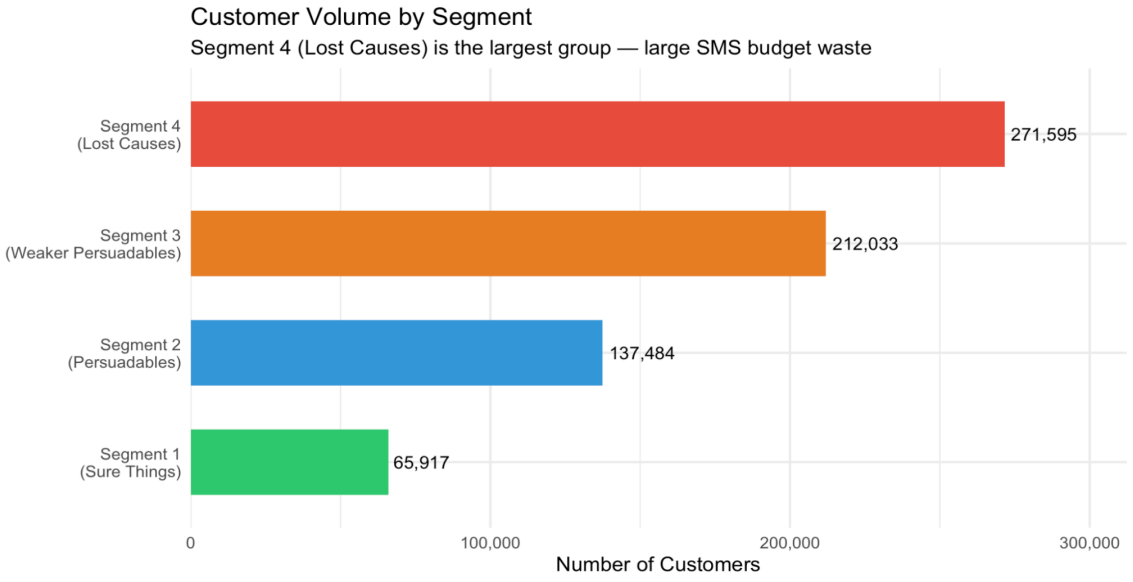

Figure 5: Customer volume by segment. Segment 4 is the largest group, while Segment 1 is the smallest.

Segment Profiles

The discovered segments form a clear behavioral gradient. Segment 1 contains the most behaviorally intense customers, with higher spending, higher receipt counts, and more frequent shopping activity. Segment 2 is still highly engaged, but not as strong as Segment 1 on baseline behavior. Segment 3 is more moderate, while Segment 4 has the weakest overall shopping intensity across the main spending and frequency variables.

The most important dimensions separating the segments are behavioral rather than demographic. Receipt counts, spending totals, and purchase frequency consistently define the cluster structure. This ordering matters because it helps explain why the strongest baseline customers are not necessarily the strongest incremental targets. A high-activity customer may still be a poor use of budget if SMS does not meaningfully change that customer's probability of visiting.

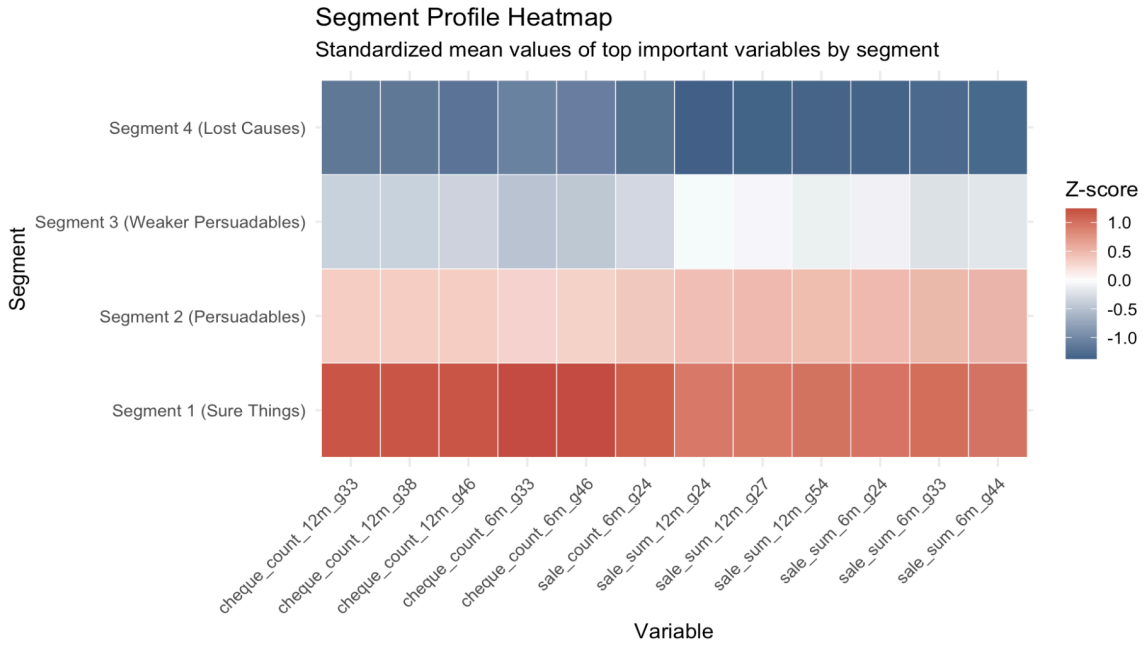

Figure 6: Segment profile heatmap. Standardized means show a strong ordering from highly engaged customers in Segment 1 to weakly engaged customers in Segment 4.

Uplift Analysis by Segment

Uplift was measured as the treated visit rate minus the control visit rate within each segment. This is the core business metric of the project because it estimates how much additional store traffic is associated with SMS rather than simply describing who visits most often.

Segment 2 produced the largest incremental lift at approximately 0.00795, or about 0.80 percentage points, making it the strongest persuadable segment. Segment 3 also responded positively, with an uplift of about 0.00586, or roughly 0.59 percentage points. Segment 1 had the highest overall visit rate, but the incremental gain from the SMS was relatively small at about 0.51 percentage points. Segment 4 showed the smallest list at about 0.43 percentage points and also had very low overall visit rates, which makes it the weakest strategic target despite its large size.

These results show why response rate alone is not enough for campaign planning. Customers with high purchase activity are not automatically the best marketing targets if the message does not materially change behavior. The most valuable customers to contact are the ones whose visits are caused by contact, not the ones who simply have high baseline visit rates.

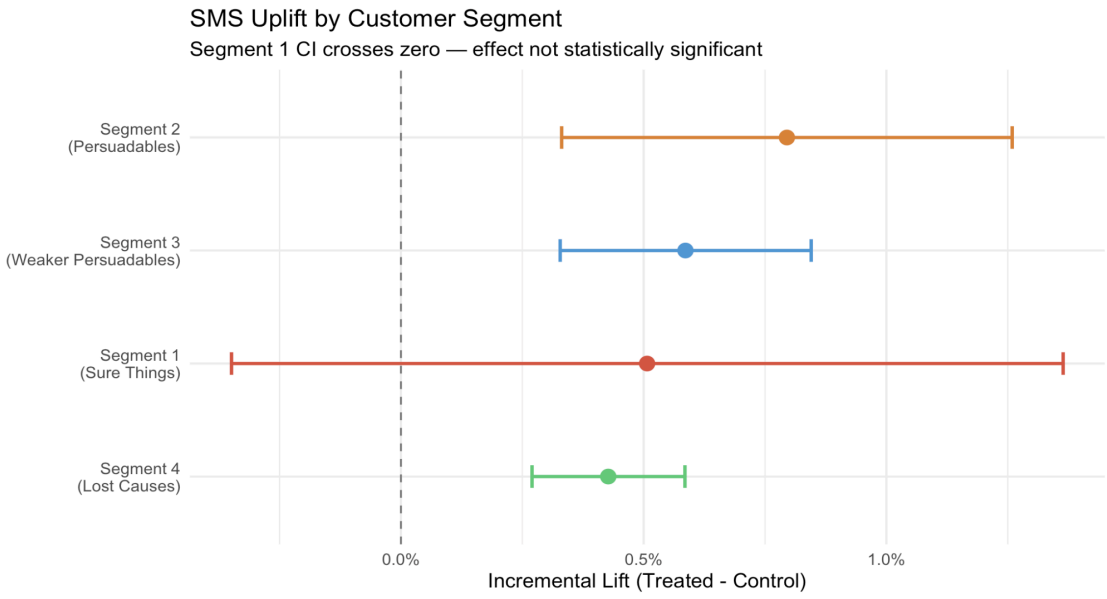

Figure 7: SMS uplift by customer segment. Segment 2 has the strongest incremental effect, followed by Segment 3.

Statistical Significance of Uplift

Confidence intervals were calculated for the segment-level uplift estimates to assess whether the observed differences were likely to reflect real treatment effects rather than random variation. Segment 2 had a 95% confidence interval of [0.00331, 0.01259], which lies entirely above zero and supports a statistically reliable positive SMS effect. Segment 3 also had a fully positive confidence interval of [0.00328, 0.00845], confirming that it is another meaningful target group.

Segment 1 had an estimated uplift of about 0.00507, but its 95% confidence interval of [-0.00349, 0.01364] crossed zero. That means the positive point estimate is not statistically distinguishable from no effect. Segment 4

had a positive interval of [0.00270, 0.00585], so the lift is likely real, but the segment's low absolute visit rates still limit its business attractiveness relative to the persuadable groups.

The key takeaway is that statistical significance and business significance are related but not identical. A segment can have positive lift and still be a weak strategic target if the baseline response remains low or if the budget opportunity cost is too high.

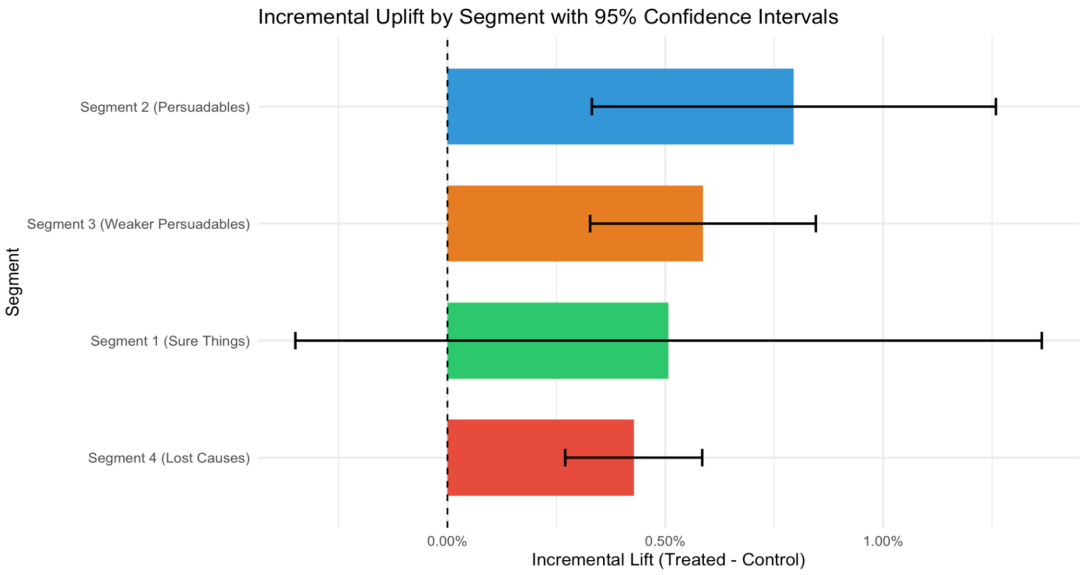

Figure 8: Incremental uplift by segment with 95% confidence intervals. Segment 1 crosses zero; Segments 2, 3, and 4 remain above zero.

Segment Interpretation in Marketing Terms

The segment-level uplift results align well with the marketing framework introduced at the beginning of the project. Segment 1 is best interpreted as Sure Things because these customers already visit at high rates and SMS does not produce a statistically reliable marginal increase. They are valuable customers, but not necessarily efficient targets for incremental outreach.

Segment 2 is the clearest Persuadables group because it combines meaningful behavioral engagement with the strongest statistically supported lift. Segment 3 can be interpreted as Weaker Persuadables because it also responds positively to SMS, but with lower baseline engagement and weaker lift than Segment 2. Segment 4 is the closest to Lost Causes because its members are weakly engaged overall and remain difficult to influence even though the estimated effect of SMS is slightly positive.

Importantly, the final results do not reveal a distinct Do-Not-Disturb segment. That absence does not undermine the framework. Instead, it suggests that in this dataset the most actionable distinction is between customers who are incrementally responsive and customers who are not.

Segment Validation with a 4-class Predictive Model

To test whether the discovered segments represented stable, learnable structure rather than arbitrary cluster assignments, a 4-class random forest model was trained to predict segment membership from customer behavior. The model was trained on 150,000 customers and 188 variables with 5-fold cross-validation. The best-performing tuning settings were mtry = 2 and splitrule = extratrees, which suggests that the feature space contains substantial redundancy and benefits from smaller candidate sets at each split.

The resulting model achieved an accuracy of 0.9522 with a confidence interval of approximately [0.9510, 0.9533], far above the no-information rate of 0.3953. Kappa was 0.9312, which indicates very strong information performance even after accounting for class imbalance. These results show that the segments are behaviorally distinct and highly reproducible.

The confusion matrix shows that most classification errors occur between neighboring groups rather than between distant ones. Segment 4 was especially easy to identify, with sensitivity of 0.9919. Segment 1 was the hardest segment to identify, with sensitivity of 0.8625, and was most often confused with Segment 2. Segment 2 and Segment 3 were also sometimes confused with one another, which supports the idea that they occupy adjacent positions on a persuadability spectrum.

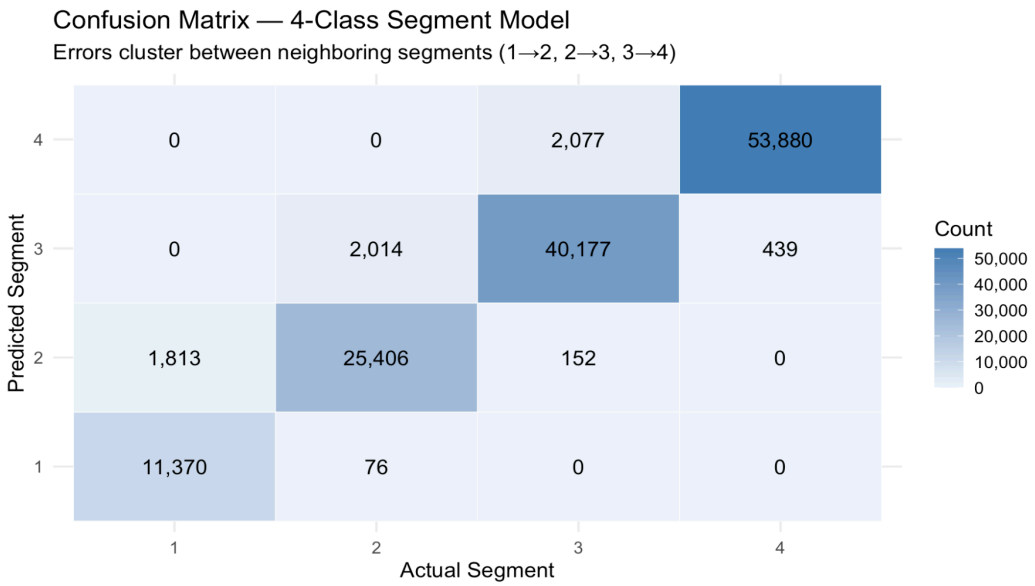

Figure 9: Confusion matrix for the 4-class segment model. Most errors occur between neighboring segments rather than distance ones.

Persuadables Model

Due to Segments 2 and 3 both having shown positive and statistically supported uplift, they were combined into a single persuadable class for a binary classification problem. Segment 1 and Segment 4 were grouped into the non-priority class because they offered weaker incremental targeting value. This binary framing is more operationally useful than the original 4-class setup, because real campaign teams often need a direct recommendation about who should be targeted first.

A binary random forest model was trained using the cleaned behavioral feature set. Performance was extremely strong, with an accuracy of 0.9682, AUC of 0.9967, and Kappa of 0.9363. The model correctly identified 68,322 persuadables, produced 2,791 false positives, and missed 1,581 persuadables. Sensitivity was 0.9774, specificity was 0.9587, and positive predictive value was 0.9608.,

These results mean the model can distinguish persuadable customers from others with very high reliability. In business terms, it translates the segmentation findings into a practical targeting tool that can support future SMS campaign selection under tight budget constraints.

Key Drivers of Persuadability

The most important predictors of segment membership and persuadability were behavioral variables tied to spending, receipt counts, purchase activity, and discount interaction. Across the models, cheque_count variables captured how often customers appeared in receipts, sale_sum variables captured spending intensity, sale_count variables captured item-level buying activity, and discount-related features reflected interaction with promotions.

This result suggests that persuadability is driven more by how customers behave than by who they are demographically. Customers with moderate-to-strong prior engagement, regular purchase behavior, and meaningful transaction activity are more likely to fall into the persuadable groups. That aligns with marketing intuition: the best targets are often not the highest-volume buyers or the least engaged customers, but customers in the middle whose behavior can still be influenced by a timely message.

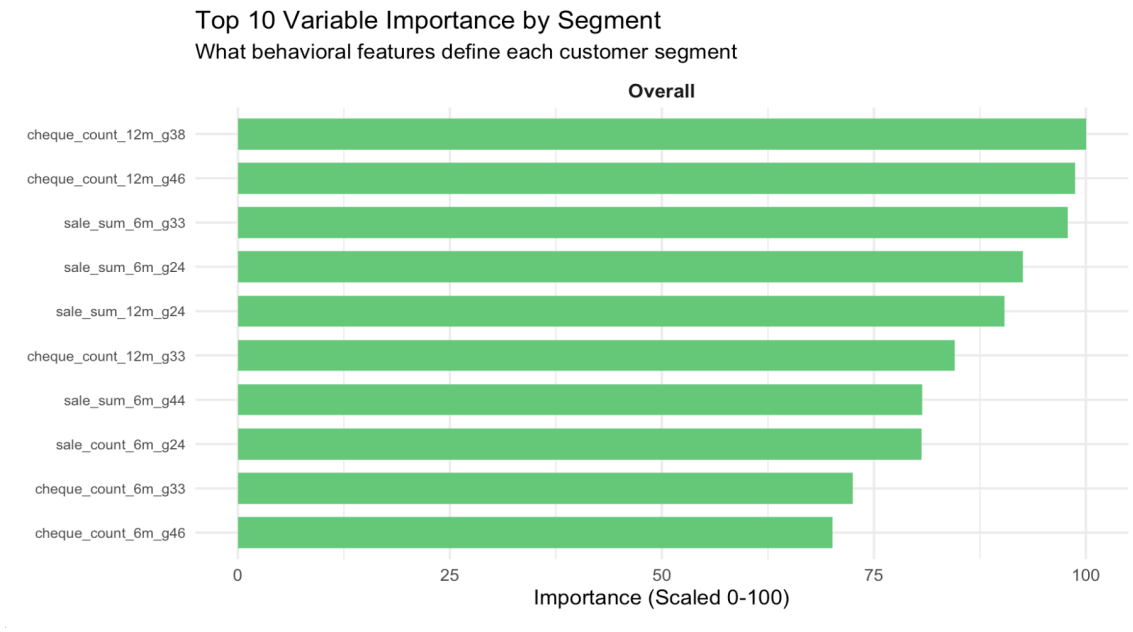

Figure 10: Variable importance for segment membership. Frequency and spending features dominate the predictive signal.

Business Recommendations

First, the company should prioritize Segment 2 for SMS targeting because it offers the strongest combination of statistical significance and incremental lift. Second, Segment 3 should be treated as a secondary target, especially when campaign capacity extends beyond the highest-priority group. Third, Segment 1 should not be a top incremental target because those customers appear likely to visit regardless of outreach, which reduces marginal return on spend.

Segment 4 should be approached cautiously. It is the largest segment, so broad messaging here can consume a large share of budget, yet its low baseline response makes it a weak strategic investment relative to the persuadable segments. A more efficient strategy is to use the binary persuadables model to rank customers and concentrate SMS volume on the highest -scoring individuals.

The broad managerial lesson is that better targeting is not about increasing message volume. It is about increasing the number of visits that happen because of the message.

Professional Relevance

This project is professionally relevant because it addresses a real CRM decision problem: how to allocate customer communication under a limited marketing budget. It demonstrates the use of data science not just for prediction, but for causal targeting, campaign optimization, and decision support.

Technically, the project combined data cleaning, feature filtering, transformation, PCA, clustering, random forests, and performance evaluation. Strategically, it applies marketing reasoning about incrementality and customer value. That combination makes the work directly relevant to CRM analytics, lifecycle marketing, business analytics, and applied data science roles.

Limitations

Several limitations should be acknowledged. First, the project estimated treatment effects at the segment level rather than using a full individual-level uplift modeling framework. Segment-level uplift is useful and interpretable, but it is still a coarser tool than direct individual treatment-effect estimation.

Second, the segment labels are interpretations based on behavior and uplift rather than permanent customer identities. Third, the final segmentation did not isolate a distinct Do-Not-Disturb group, so the empirical results only partially map onto the full theoretical four-group framework. Fourth, some variables were removed because of sparsity or missingness, which means potentially useful information may have been excluded. Finally, the analysis focused on incremental visits rather than full financial return, so it does not directly optimize profit, margin, or lifetime value.

Future Work

A natural next step would be to build an individual-level uplift model that ranks customers directly by expected treatment effect rather than assigning lift at the segment level. A further improvement would be to incorporate SMS cost, order value, and customer lifetime value so that targeting decisions optimize profit rather than only visit lift.

The segmentation itself could also be refreshed over time to capture changes in customer behavior. In practice, a live field test using the persuadables model would be especially valuable, because it would allow the company to compare targeted deployment against the current campaign strategy and measure business impact in production.

Conclusion

This capstone set out to answer a clear business question: if SMS budget is limited, how can customers be targeted to maximize incremental store visits relative to sending no SMS? The results show that customers are not equally responsive to SMS and that behavior-based segmentation provides a useful way to separate high-value targets from low-value ones.

Segment 2 emerged as the strongest persuadable group, Segment 3 also responded positively, Segment 1 behaved like a Sure Things group, and Segment 4 functioned as the weakest strategic target despite being the largest

segment. The most important lesson is that effective CRM targeting depends on identifying customers whose behavior is changed by contact, not simply customers who are likely to visit.

Under a limited budget, the best SMS strategy is not to contact more customers, but to contact the customers whose visits are truly incremental.

Appendix: Additional Suggested Figures

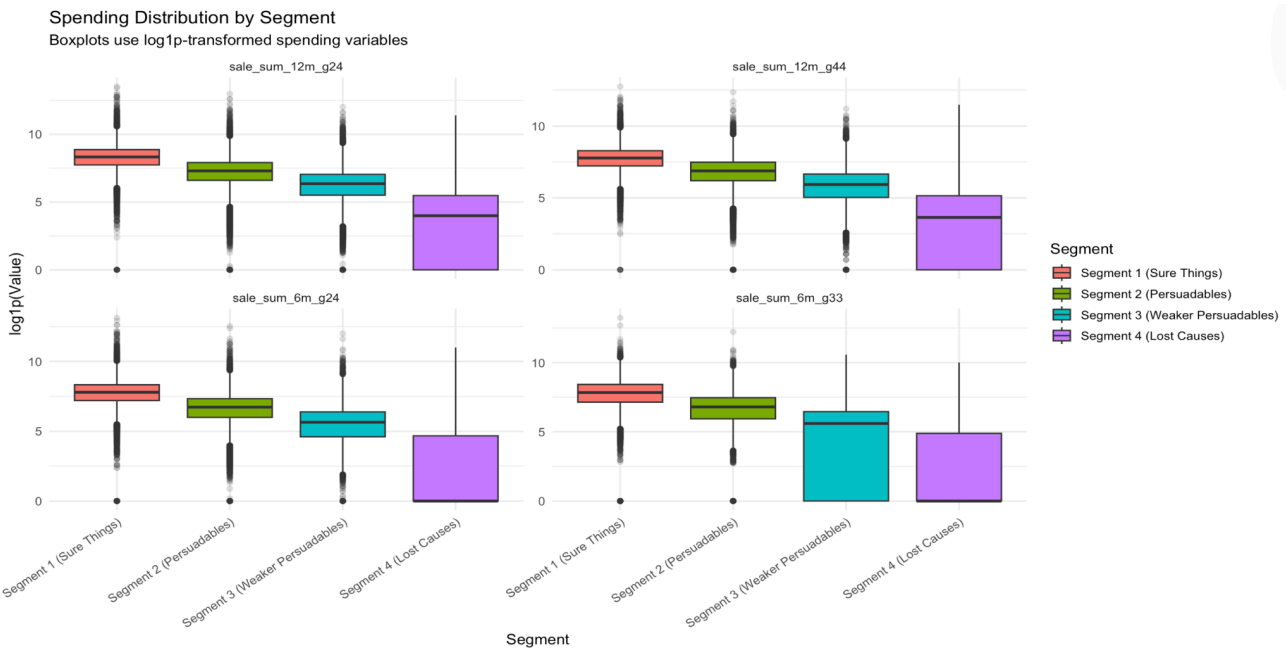

Figure A1: Spending distributions by segment. Spending intensity declines across segments, supporting the interpretation that the groups differ meaningfully in customer value and engagement.

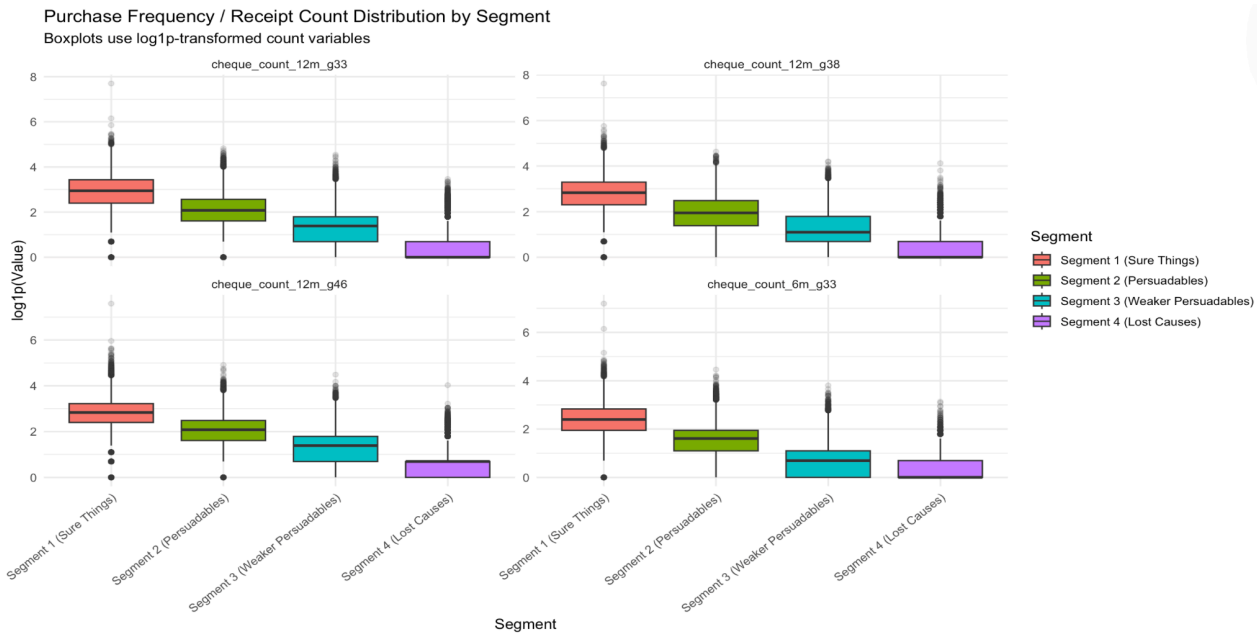

Figure A2: Purchase frequency and receipt count distributions by segment. Higher-activity segments show consistently greater receipt volume, reinforcing the behavioral ordering from Sure Things to Lost Causes.

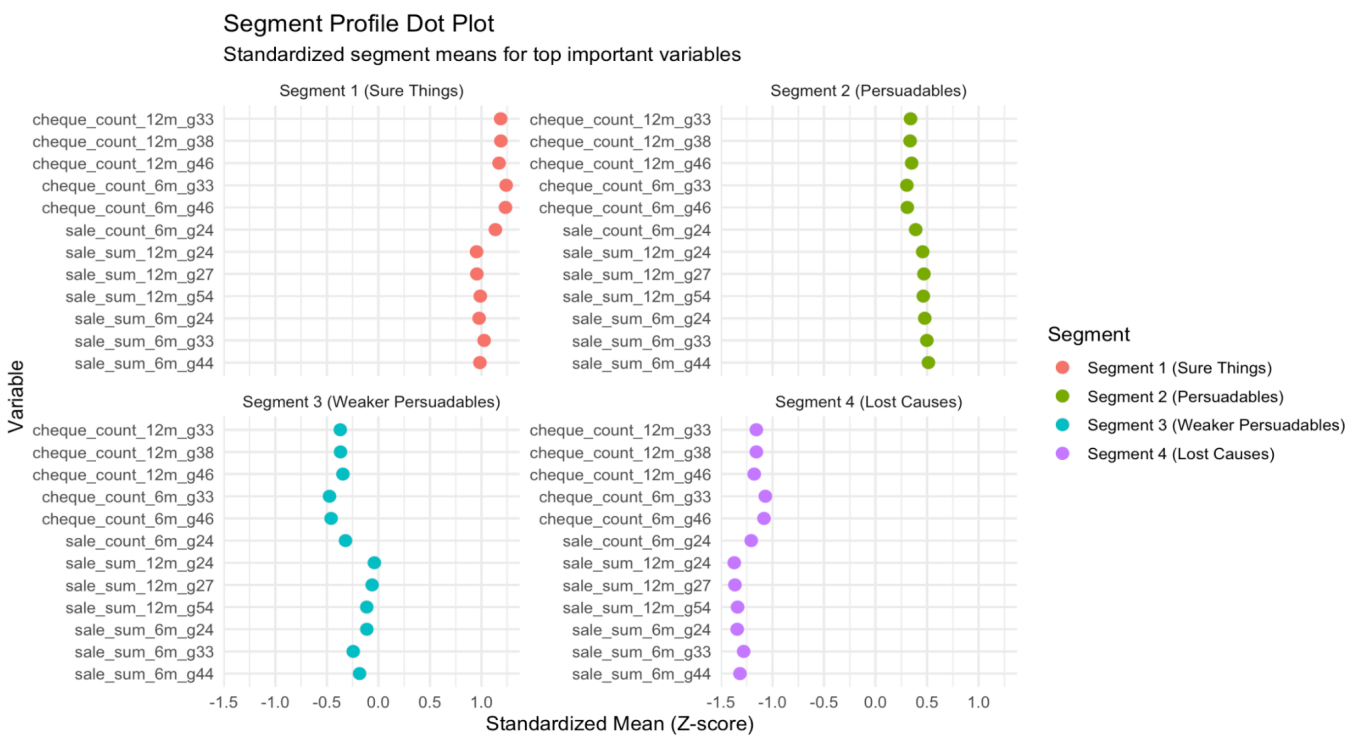

Figure A3: Standardized segment profile dot plot. Variable-by-variable comparisons show that frequency and spending features separate the four segments most clearly.

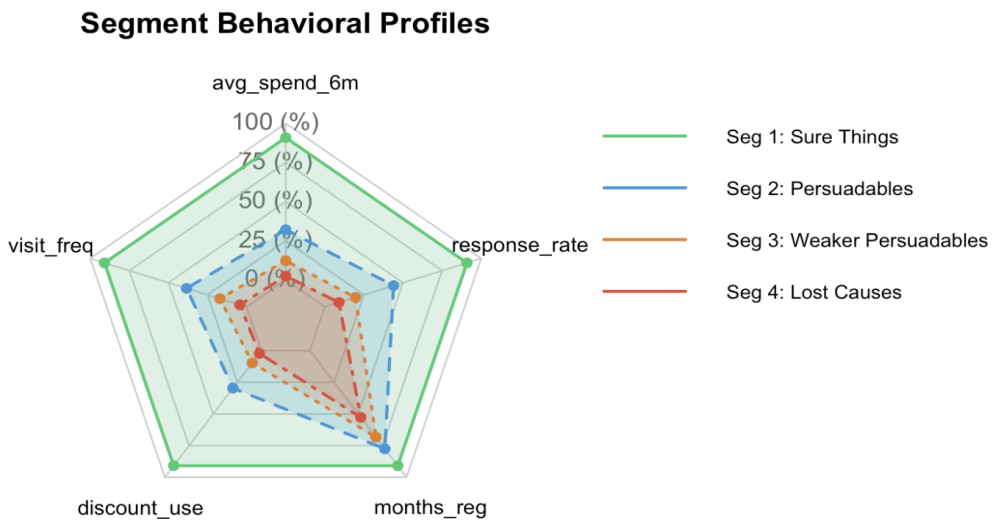

Figure A4: Radar chart of segment behavioral profiles. Broad behavioral dimensions highlight the strong engagement of Sure Things, the moderate engagement of Persuadables, and the weaker activity of Lost Causes.